

Klastoets 2
VRAESTEL

Kopiereg voorbehou

Class Test 2
QUESTION PAPER

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Module EBN111
Elektrisiteit en Elektronika
April 2018

Module EBN111
Electricity and Electronics
April 2018



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:
Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form 30 min + 10 min om OKH vorm te voltooi	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		8 + OCR	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (0-5 min)

Do step 1 in the first 30 minutes of the test. (5-35 min)

Do step 2 within the following 10 minutes. (35-45 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111C2**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell boundaries.
- **Do not use commas!** Please make use of full stops.

Examples

01	-	1	.	4	5					m	A						
02	3	4	0							v							
03	1	5	.	2													
04	-	1								Ω							

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die sessie. (0-5 min)

Doen stap 1 in die eerste 30 minute van die toets. (5-35 min)

Doen stap 2 binne die daaropvolgende 10 minute. (35-45 min)

STAP 0 (-5 min to 0 min): Voltooi die voorblad van di OKH vorm

STAP 1 (0 min to 30 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

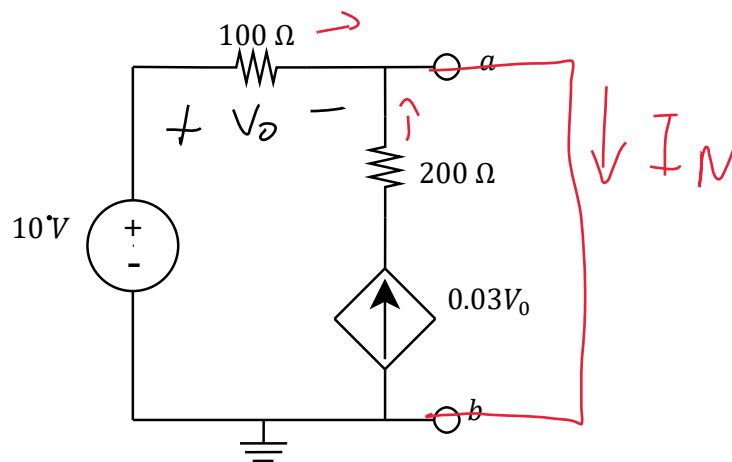
STAP 2 (35 min to 40 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

- Die "Assessment ID" is **2018EBN111C2**
- Gebruik 'n meganiese potlood met 0.5mm HB lood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerder gebruik van punte.

Voorbeelde

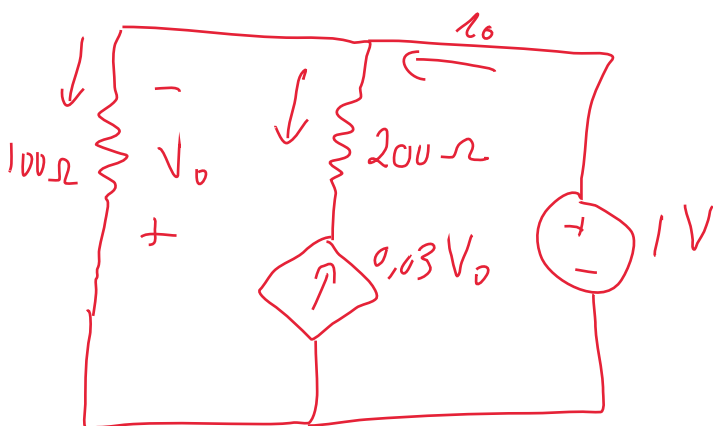
01	-	1	.	4	5					m	A						
02	3	4	0							v							
03	1	5	.	2													
04	-	1								Ω							

Questions 1-2 [4]		Vraag 1-2 [4]	
Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals $a-b$.		Bepaal die Norton ekwivalente stroom I_N en weerstand R_N met verwysing tot terminale a en b .	
01	I_N	01	I_N
02	R_N	02	R_N



$$I_N = \frac{10}{100} + 0,03(10)$$

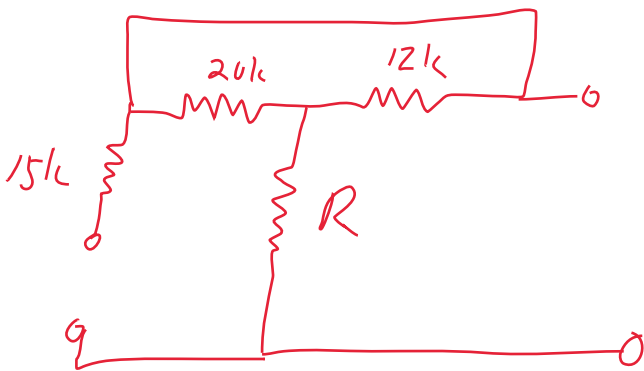
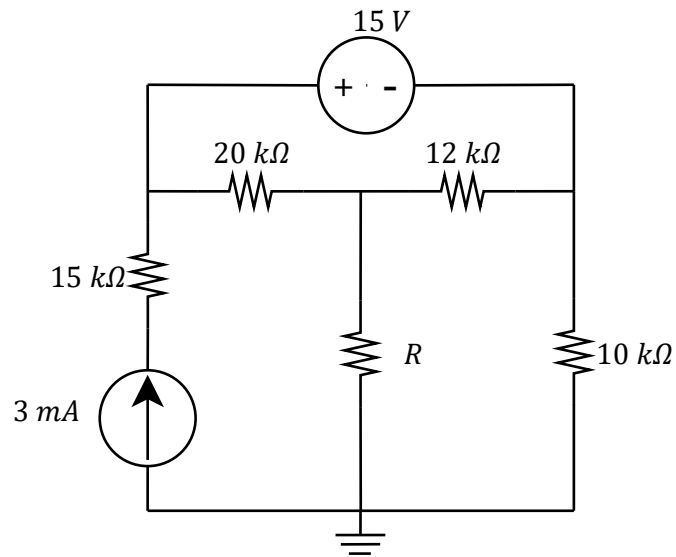
$$I_N = 0,4 \text{ A}$$



$$\begin{aligned} i_0 &= \frac{1}{100} - 0,03V_0 \\ &= 0,01 + 0,03 \\ &= 0,04 \text{ A} \end{aligned}$$

$$\therefore R_N = \frac{1}{0,04} = 25 \Omega$$

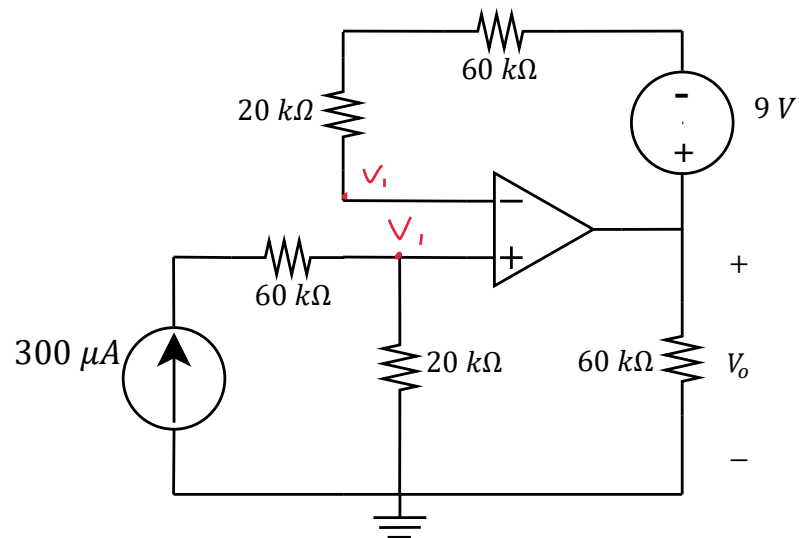
Questions 3 [2]		Vraag 3 [2]	
03	Find R such that maximum power is delivered to the $10\text{ k}\Omega$ resistor.	03	Bepaal R sodat maksimum drywing aan die $10\text{ k}\Omega$ resistor oorgedra word.



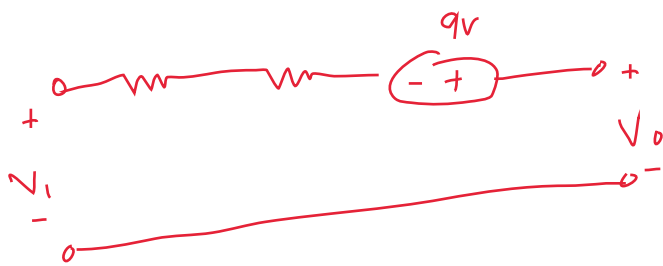
$$(20k // 12k) + R = 10k$$

$$\underline{R = 2,5k\Omega}$$

Question 4 [2]		Vraag 4 [2]	
04	Determine V_o .	04	Bepaal V_o .



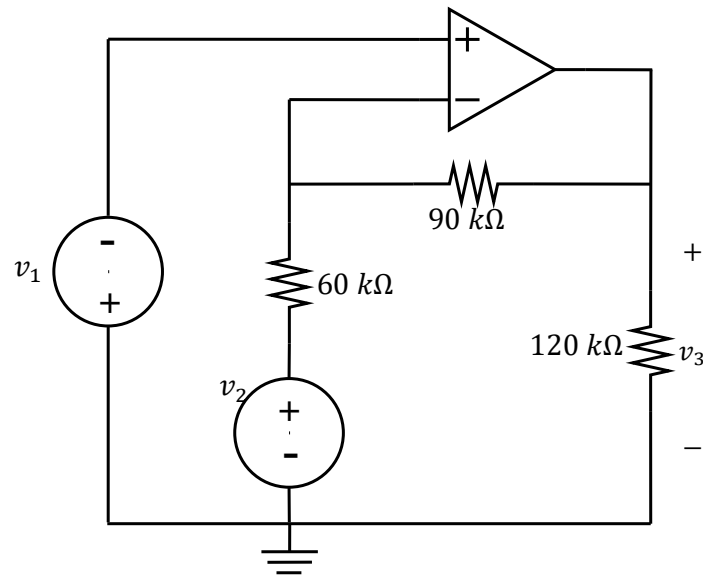
$$V_i = (20k)(300\mu A) = 6V$$



$$-V_i - 9 + V_o = 0$$

$$\therefore V_o = 15V$$

Questions 5-6 [4]		Vraag 5-6 [4]	
05	Determine v_2 if $v_3 = -12V$ and $v_1 = 0V$.	05	Bepaal v_2 indien $v_3 = -12V$ en $v_1 = 0V$.
06	Determine v_3 if $v_1 = 3V$ and $v_2 = 0V$.	06	Bepaal v_3 indien $v_1 = 3V$ en $v_2 = 0V$.



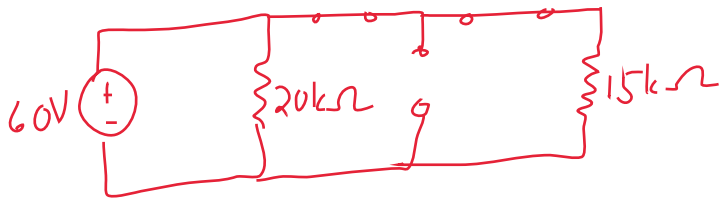
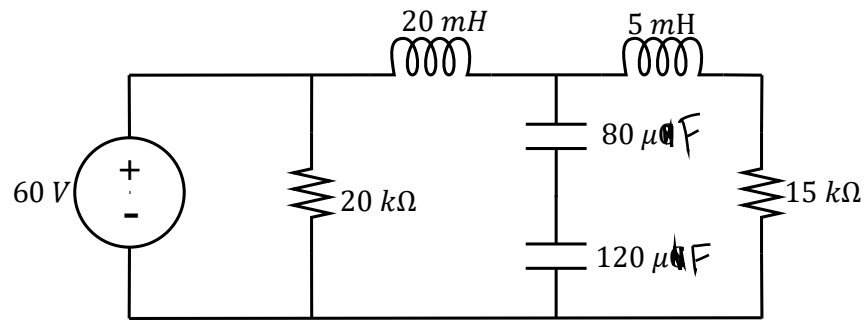
$$V_3 = -\frac{90k}{60k} V_2 \Rightarrow V_2 = -\frac{6}{9}(-12)$$

$$\underline{V_2 = 8V}$$

$$V_3 = \left(1 + \frac{90k}{60k}\right)(-3)$$

$$\underline{V_3 = -7,5V}$$

Questions 7 [2]		Vraag 7 [2]	
07	For the DC circuit below, determine the total energy w_C stored in the capacitors.	07	Indien GS-toestand aanvaar word, bepaal die totale energie w_C gestoor in die kapasitors.

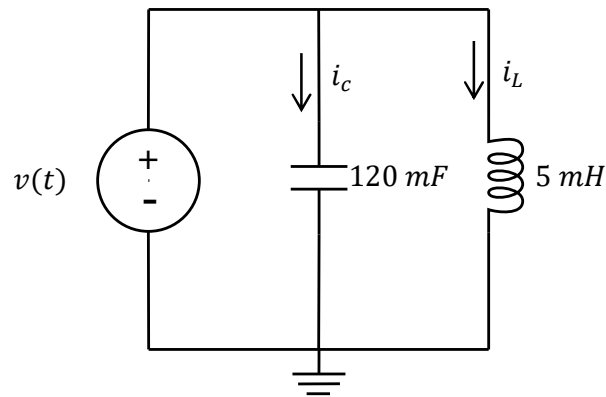


$$w_C = \frac{1}{2} C_{eq} V^2 = \frac{1}{2} (80 \times 10^{-6} // 120 \times 10^{-6}) (60)^2$$

$$= \frac{1}{2} (48 \times 10^{-6}) (3600)$$

$$w_C = 86,4 \text{ mJ}$$

Questions 8 [2]		Vraag 8 [2]	
Determine $i_C(2ms)$ if $v(t) = 20 \cos(50\pi t) mV$.		Bepaal $i_C(2ms)$ indien $v(t) = 20 \cos(50\pi t) mV$.	
08	$i_C(2ms)$	08	$i_C(2ms)$



$$i_c(t) = (120m) \frac{d}{dt} (20m \cos(50\pi t))$$

$$= 2,4m (-50\pi) \sin(50\pi t)$$

$$i_c(2m) = -0,12\pi \sin(50\pi(2m))$$

$$\underline{i_c(2m) = -116,4967m A} \rightarrow$$

PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	1	1	C	2		
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Answer

Unit

01	$I_N = 0,4$	A
02	$R_N = 25$	Ω
03	$R = 2,5$	$k\Omega$
04	$V_o = 15$	V
05	$v_2 = 8$	V
06	$v_3 = -7,5$	V
07	$w_C = 86,4$	mJ
08	$i_C(2ms) = -116,4967$	mA